UI/UX DESIGN SPECIFICATION

1. Product

AI-Powered Integrated Bid Compliance Verification Platform

SIH 2026 Problem Statement: 26100

---

2. UI/UX OBJECTIVE

Design a professional, modern procurement/compliance platform that makes a complex verification process easy to understand.

The interface must communicate:

What was required → What was submitted → What was extracted → What was verified → What passed/failed → Why → What needs human review

The UI should feel like a serious enterprise/government procurement application, not a generic AI chatbot.

---

3. DESIGN PRINCIPLES

Follow these principles:

1. **Clarity over decoration**
2. **Evidence before AI conclusions**
3. **Role-based simplicity**
4. **Progressive disclosure**
5. **Consistent navigation**
6. **Strong visual hierarchy**
7. **Accessible design**
8. **Responsive layout**
9. **Security-conscious UX**
10. **Human-in-the-loop visibility**

Do not overwhelm users with technical AI information unless they need it.

---

4. DESIGN STYLE

Use a modern enterprise SaaS visual language.

Overall Feel

- Professional
- Clean
- Trustworthy
- Minimal
- Data-oriented
- Government/enterprise appropriate
- Modern but not flashy

Avoid:

- Excessive gradients
- Excessive animations
- Glassmorphism everywhere
- Huge decorative illustrations
- Neon colors
- Gaming-style interfaces

---

5. COLOR SYSTEM

Use a neutral enterprise palette.

Primary

Use a deep professional blue/indigo tone for:

- Primary buttons
- Active navigation
- Links
- Important actions

Success

Use green for:

- Compliant
- Verified
- Completed
- Successful processing

Warning

Use amber/yellow for:

- Needs review
- Low confidence
- Pending
- Verification unavailable

Error

Use red for:

- Non-compliant
- Failed
- Rejected
- Security errors

Neutral

Use:

- White
- Light gray
- Medium gray
- Dark text
- Dark surfaces where required

Do not use color alone to communicate status.

Always combine:

Color + Icon + Text

Example:

```
✓ COMPLIANT
⚠ NEEDS REVIEW
× NON-COMPLIANT
```

---

6. TYPOGRAPHY

Use a clean modern font such as:

- Inter
- Geist
- system sans-serif

Recommended hierarchy:

```
Page Title
    ↓
Section Heading
    ↓
Card Heading
    ↓
```

```
Body Text
    ↓
Secondary Text
```

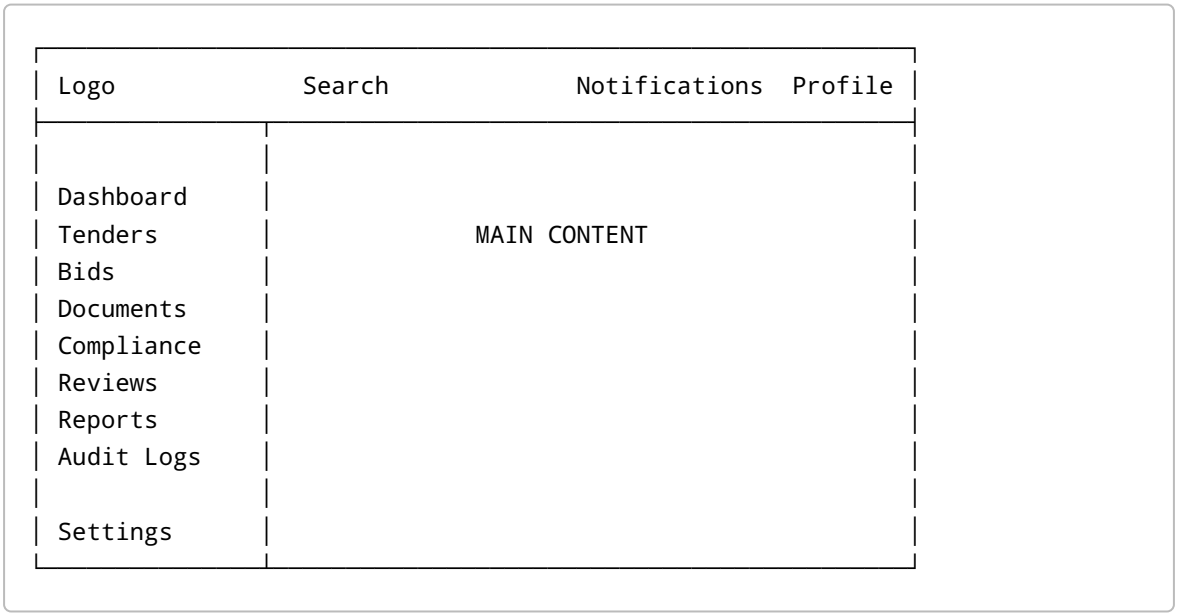
Maintain strong readability.

---

7. LAYOUT

Desktop-first enterprise dashboard.

Recommended structure:



Sidebar should be collapsible.

---

8. RESPONSIVE DESIGN

Support:

- Desktop
- Laptop
- Tablet
- Mobile

For mobile:

- Collapse sidebar
- Use bottom/action menus where appropriate
- Convert large tables into cards
- Preserve critical actions

- Allow document preview
- Maintain readable compliance statuses

---

9. GLOBAL NAVIGATION

Navigation should change according to role.

Admin

```
Dashboard
Users
Roles & Permissions
Tenders
Bids
Reviews
Reports
Audit Logs
Verification Providers
Settings
```

Procurement Officer

```
Dashboard
Tenders
Bids
Compliance
Reviews
Reports
```

Bidder

```
Dashboard
Tenders
My Bids
Documents
Compliance
Requests
Profile
```

Reviewer

```
Dashboard
Review Queue
```

```
Assigned Reviews
Compliance
Reports
```

Auditor

```
Dashboard
Audit Logs
Verification History
Decision History
Reports
```

---

10. LOGIN PAGE

Layout

Centered authentication card.

```
┌──────────────────────────────────────┐
│                                      │
│          [LOGO]                      │
│                                      │
│     Bid Compliance Platform          │
│                                      │
│  Email                               │
│  [____________________________]      │
│                                      │
│  Password                            │
│  [____________________________]      │
│                                      │
│  [        Sign In        ]           │
│                                      │
│  Forgot Password?                    │
│                                      │
└──────────────────────────────────────┘
```

Show:

- Secure connection indication where appropriate
- Validation errors
- Loading state
- Authentication failure message

Never display technical authentication errors.

---

11. GLOBAL HEADER

Header contains:

Left

- Logo
- Product name
- Breadcrumb

Center

Optional global search.

Right

- Notifications
- Help
- User profile
- Role indicator

Example:

```
BidVerify
  / Tenders / Tender #TND-1024

                          🔔   👤 Officer
```

---

12. DASHBOARD

Dashboard should provide an immediate overview.

Officer Dashboard

Top KPI cards:

```
┌──────────────┐ ┌──────────────┐ ┌──────────────┐ ┌──────────────┐
│  Tenders     │ │    Bids      │ │  Reviews     │ │  Issues      │
│     24       │ │    186       │ │     17       │ │     12       │
└──────────────┘ └──────────────┘ └──────────────┘ └──────────────┘
```

Then:

- Active tenders
- Recent bids

- Compliance distribution
- Pending reviews
- Verification failures
- Recent activity

---

13. BIDDER DASHBOARD

Display:

```
My Bids
_____________________________________

Tender              Status

Medical Equipment   ✓ Compliant
IT Services         ⚠ Under Review
Security Service    ⏳ Processing
```

Additional cards:

- Active bids
- Documents uploaded
- Missing documents
- Pending requests
- Completed submissions

---

14. TENDER LIST PAGE

Use a searchable/filterable table.

Columns:

```
Tender ID
Title
Organization
Deadline
Status
Bids
Actions
```

Example:

```
TND-1024
Medical Equipment Procurement
Government Organization
12 Sep 2026
Published
18 Bids
View
```

Filters:

- Status
- Category
- Date
- Organization

---

15. CREATE TENDER PAGE

Use a multi-step wizard.

```
1. Basic Details
      ↓
2. Tender Document
      ↓
3. Requirements
      ↓
4. Review
      ↓
5. Publish
```

Step 1

Fields:

- Tender title
- Tender ID
- Organization
- Category
- Description
- Submission deadline

Step 2

Upload tender document.

Step 3

Display extracted requirements.

Allow officer to:

- Edit
- Add
- Delete
- Mark mandatory
- Change category

Step 4

Show final preview.

Step 5

Publish confirmation.

---

16. TENDER DETAILS PAGE

Structure:

```
Tender Header
_____________________________________

Tender ID
Title
Organization
Deadline
Status

[Edit] [View Requirements] [View Bids]

_____________________________________

Overview
Requirements
Documents
Bids
Activity
```

---

17. REQUIREMENT VIEW

Requirements should be displayed as structured cards/table.

Example:

```
Requirement #REQ-001

Category
Financial

Requirement
Minimum annual turnover of ₹50 lakh

Mandatory
YES

Threshold
₹50,00,000

Source
Tender.pdf — Page 12

Confidence
94%
```

Actions:

- Edit
- Validate
- View source page

---

18. BID SUBMISSION

Bidder should have a guided submission workflow.

```
Tender Details
      ↓
Bid Information
      ↓
Required Documents
      ↓
Upload
      ↓
Validation
      ↓
```

```
Review
      ↓
Submit
```

Do not force users to navigate multiple unrelated pages.

---

19. DOCUMENT UPLOAD UI

Use drag-and-drop plus normal file picker.

```
┌──────────────────────────────────────────┐
│                                          │
│        Drag & Drop Documents             │
│                                          │
│              or                          │
│                                          │
│          [ Browse Files ]                │
│                                          │
│ PDF, DOCX, JPG, PNG                      │
│ Maximum size: configured by system       │
└──────────────────────────────────────────┘
```

After upload:

```
GST Certificate.pdf
████████████████ 100%
✓ Uploaded
```

Processing:

```
GST Certificate.pdf

✓ Uploaded
✓ Validated
✓ OCR Completed
⏳ Extracting Information
```

---

20. DOCUMENT MANAGEMENT PAGE

Display:

```
Document
Type
Upload Date
Processing Status
Verification
Actions
```

Example:

```
GST Certificate
Certificate
27 Aug 2026
✓ Processed
✓ Verified
View
```

Allow:

- Preview
- Download where authorized
- View extracted data
- View processing status

---

21. DOCUMENT VIEWER

Use a split-screen layout.

```
┌──────────────────────┬──────────────────────────┐
│                      │                          │
│   DOCUMENT PREVIEW   │   EXTRACTED DATA         │
│                      │                          │
│   Page 4             │   Company Name           │
│                      │   ABC Technologies       │
│                      │                          │
│                      │   GSTIN                  │
│                      │   XXXXX                  │
│                      │                          │
│                      │   Confidence: 96%        │
│                      │                          │
└──────────────────────┴──────────────────────────┘
```

Where possible, clicking extracted evidence should highlight its source location.

---

22. AI PROCESSING STATUS

Never make AI processing appear instant if it is actually asynchronous.

Show stages:

```
Document received
       ↓
Validating
       ↓
OCR processing
       ↓
Extracting information
       ↓
Matching requirements
       ↓
Running verification
       ↓
Compliance analysis
       ↓
Completed
```

Use progress indicators and status text.

---

23. COMPLIANCE DASHBOARD

This is the most important screen.

Top section:

```
Overall Compliance

        82%
     COMPLIANT

18 Requirements
14 Passed
2 Failed
2 Review
```

Then requirement-level analysis.

---

24. REQUIREMENT COMPLIANCE CARD

Example:

```
┌──────────────────────────────────────────────────┐
│ ✓ COMPLIANT                                      │
│                                                  │
│ REQ-001 — Minimum Annual Turnover                │
│                                                  │
│ Required: ₹50 lakh                               │
│ Extracted: ₹72 lakh                              │
│                                                  │
│ Evidence: Financial Statement                    │
│ Page: 4                                          │
│                                                  │
│ Confidence: 94%                                  │
│ Verification: Verified                           │
│                                                  │
│ [View Evidence] [View Details]                   │
└──────────────────────────────────────────────────┘
```

---

25. NON-COMPLIANT CARD

```
┌──────────────────────────────────────────────────┐
│ × NON-COMPLIANT                                  │
│                                                  │
│ REQ-007 — Minimum Experience                     │
│                                                  │
│ Required: 5 years                                │
│ Extracted: 2 years                               │
│                                                  │
│ Evidence: Experience Certificate — Page 3        │
│                                                  │
│ Reason                                           │
│ Submitted experience is below the required       │
│ threshold.                                       │
│                                                  │
│ [View Evidence]                                  │
└──────────────────────────────────────────────────┘
```

---

26. NEEDS REVIEW CARD

```
┌──────────────────────────────────────────────────┐
│ ⚠ NEEDS REVIEW                                   │
│                                                  │
│ REQ-012 — Organization Name                      │
│                                                  │
│ GST: ABC Technologies Pvt Ltd                    │
│ PAN: ABC Technology Pvt Ltd                      │
│                                                  │
│ Potential mismatch detected.                     │
│                                                  │
│ Confidence: 61%                                  │
│                                                  │
│ [Review Case]                                    │
└──────────────────────────────────────────────────┘
```

---

27. COMPLIANCE FILTERS

Allow users to filter:

- All
- Compliant
- Non-compliant
- Needs Review
- Missing Evidence
- Verification Failed

Additional filters:

- Requirement category
- Confidence
- Document type
- Verification status

---

28. COMPLIANCE DETAIL PAGE

Use a three-column conceptual structure:

```
Requirement
     │
     ▼
Evidence
```

```
     |
     ▼
Decision
```

UI:

```
┌──────────────────┬──────────────────────┬──────────────────────┐
│ REQUIREMENT      │ EVIDENCE             │ DECISION             │
│                  │                      │                      │
│ Requirement      │ Document             │ COMPLIANT            │
│ text             │ Page                 │                      │
│                  │ Extracted value      │ Confidence           │
│ Rule             │ Source evidence      │ Verification         │
└──────────────────┴──────────────────────┴──────────────────────┘
```

---

29. AI EXPLANATION UI

AI reasoning should be displayed as a concise explanation, not hidden chain-of-thought.

Example:

```
Why this result?

The tender requires a minimum turnover of ₹50 lakh.
The submitted financial document contains a turnover
value of ₹72 lakh.

Therefore, the requirement is satisfied.

Evidence
Financial Statement — Page 4
```

Do not expose internal chain-of-thought or hidden reasoning.

Show only:

- Relevant evidence
- Decision factors
- Concise explanation

---

30. VERIFICATION UI

Display verification status prominently.

Example:

```
Verification

✓ Verified
Source: Authorized Verification Provider
Reference: VER-92831
Checked: 27 Aug 2026, 15:42
```

Unavailable:

```
⚠ Verification Unavailable

The verification provider could not be reached.

This result should not be treated as verified.

[Retry] [Send for Review]
```

Mock:

```
i MOCK VERIFICATION

This result was generated using the configured
demonstration verification provider.
```

---

31. REVIEW QUEUE

Reviewer dashboard should prioritize cases.

Table:

```
Priority
Bidder
Requirement
Issue
Confidence
Age
Action
```

Example:

```
HIGH
ABC Technologies
GST Name
Potential mismatch
61%
2h
Review
```

Use sorting:

- Priority
- Confidence
- Age
- Tender
- Bidder

---

32. REVIEW WORKSPACE

Use split view.

```
┌──────────────────────┬──────────────────────────────────┐
│ Requirement          │ Evidence                         │
│                      │                                  │
│ Requirement text     │ Document viewer                  │
│                      │                                  │
│ AI Result            │ Extracted fields                 │
│ NEEDS REVIEW         │                                  │
│                      │ Verification result              │
├──────────────────────┴──────────────────────────────────┤
│ Reviewer Decision                                       │
│                                                         │
│ ○ Compliant                                             │
│ ○ Non-Compliant                                         │
│ ○ Needs Clarification                                   │
│                                                         │
│ Reason                                                  │
│ [_____________________________________________]         │
│                                                         │
│ [Save Decision]                                         │
└─────────────────────────────────────────────────────────┘
```

---

33. REVIEW OVERRIDE UX

If reviewer changes AI decision, require justification.

Example:

```
AI Decision:
NON-COMPLIANT

Reviewer Decision:
COMPLIANT

Reason:
Document page 5 contains additional evidence that
was not correctly extracted during automated processing.
```

The UI should make it clear that a human has overridden the automated result.

---

34. REPORT PAGE

Report preview:

```
Bid Compliance Report

Tender Information
_____________________________

Bidder Information
_____________________________

Compliance Summary
_____________________________

Requirement Results
_____________________________

Verification Results
_____________________________

Exceptions
_____________________________

Reviewer Decisions
_____________________________

Audit Information
```

```
_____________________________

[Generate PDF]
```

---

35. AUDIT LOG UI

Use a searchable timeline/table.

```
Timestamp
Actor
Action
Resource
Result
```

Example:

```
27 Aug 2026 15:42
Reviewer
COMPLIANCE_DECISION_CHANGED
REQ-1024
SUCCESS
```

Clicking an event opens detailed metadata according to the user's permissions.

---

36. NOTIFICATION CENTER

Notification types:

- Document processing completed
- Missing document
- Review required
- Verification completed
- Compliance completed
- Bid status changed

Example:

```
🔔 Notifications

⚠ Review required
Bid #BID-1024 has 2 requirements
requiring manual review.
```

```
✓ Processing complete
Financial Statement has been processed.
```

---

37. SEARCH

Global search should support authorized resources:

- Tender ID
- Bid ID
- Bidder name
- Document name
- Requirement ID

Search results must respect RBAC.

Never expose unauthorized records through search.

---

38. EMPTY STATES

Every page needs a meaningful empty state.

Example:

```
No pending reviews

All assigned compliance cases have been reviewed.

[View Completed Reviews]
```

Do not show blank screens.

---

39. LOADING STATES

Use skeleton loaders for:

- Dashboard
- Tables
- Compliance cards
- Document metadata

For long operations use explicit processing status.

Avoid excessive spinners.

40. ERROR STATES

Errors must be understandable.

Bad:

```
500 Internal Server Error
```

Better:

```
We couldn't process this document.

The document processing service is temporarily
unavailable.

[Retry]
```

Do not expose stack traces or internal technical information.

---

41. CONFIRMATION DIALOGS

Use confirmation dialogs for destructive/high-impact actions.

Examples:

- Delete document
- Cancel tender
- Reject bid
- Change final decision
- Disable user

Example:

```
Confirm Decision

You are changing this requirement from
NON-COMPLIANT to COMPLIANT.

Please provide a reason.

[Cancel] [Confirm Decision]
```

---

42. DOCUMENT UPLOAD SECURITY UX

When uploading:

Show accepted formats and limits.

Example:

```
Supported: PDF, DOCX, JPG, PNG
Maximum size: 20 MB

Files are securely processed.
```

If rejected:

```
File rejected

This file type is not supported.
Please upload a supported document.
```

Do not display security implementation details to normal users.

---

43. ACCESS DENIED PAGE

```
Access Restricted

You don't have permission to access this resource.

If you believe this is incorrect, contact your
administrator.

[Back to Dashboard]
```

Do not reveal whether a hidden resource exists when authorization rules require concealment.

---

44. PROFILE PAGE

Show:

- Name
- Email
- Role

- Account status
- Last login where appropriate
- Security settings

Users should not be able to change privileged roles themselves.

---

45. ADMIN USER MANAGEMENT

Table:

```
User
Email
Role
Status
Last Active
Actions
```

Actions:

- View
- Edit
- Activate
- Disable
- Assign role

High-impact actions require confirmation.

---

46. DESIGN COMPONENTS

Create reusable components:

```
Button
Input
Select
Modal
Dialog
Card
Badge
StatusBadge
DataTable
Tabs
Breadcrumb
Pagination
Dropdown
Toast
```

```
Alert
Tooltip
ProgressBar
ProgressStepper
FileUploader
DocumentViewer
EvidenceViewer
ComplianceCard
VerificationBadge
ReviewPanel
AuditTimeline
```

Do not create separate inconsistent components for the same UI purpose.

---

47. STATUS COMPONENT

Create one reusable status system.

```
COMPLIANT
NON_COMPLIANT
NEEDS_REVIEW
VERIFIED
UNVERIFIED
VERIFICATION_FAILED
PROCESSING
FAILED
PENDING
APPROVED
REJECTED
```

Each status must have:

- Label
- Icon
- Visual treatment
- Accessible text

---

48. TABLE UX

Tables should support:

- Search
- Sorting
- Filtering
- Pagination

- Row actions
- Responsive behavior

For mobile, transform important rows into cards.

Avoid extremely wide tables.

---

49. ACCESSIBILITY

Target WCAG-friendly UX.

Implement:

- Keyboard navigation
- Visible focus states
- Semantic HTML
- Accessible labels
- Screen-reader-friendly status
- Adequate text contrast
- Form validation messages
- Keyboard-accessible dialogs

Never communicate critical information using color alone.

---

50. UX FOR AI UNCERTAINTY

The system must never make AI appear more certain than it is.

Use:

```
AI Confidence: 61%
⚠ Human Review Recommended
```

instead of:

```
AI says definitely wrong
```

The UI should clearly separate:

AI recommendation

from:

Final human decision

51. UX FOR EVIDENCE

Evidence should always be one click away.

Every compliance result should provide:

```
[View Evidence]
```

Clicking it should open:

- Document
- Page
- Relevant extracted information
- Verification information

where authorized.

52. UX FOR FINAL DECISION

Final decision page should summarize:

```
FINAL BID STATUS

COMPLIANT

Requirements:
18

Passed:
16

Failed:
1

Needs Review:
1

Verified:
14

Unverified:
2
```

```
Human Reviewed:
3
```

If unresolved issues exist, make them prominent.

---

53. FRONTEND ROUTE STRUCTURE

Suggested:

```
/login

/dashboard

/tenders
/tenders/new
/tenders/:id
/tenders/:id/requirements
/tenders/:id/bids

/bids
/bids/:id
/bids/:id/documents
/bids/:id/compliance
/bids/:id/report

/documents
/documents/:id

/reviews
/reviews/:id

/reports

/audit

/users
/settings
/profile
```

Routes must be protected according to role and resource authorization.

---

54. FRONTEND STATE ARCHITECTURE

Separate:

Server State

- Tenders
- Bids
- Documents
- Compliance
- Reviews
- Reports

UI State

- Modal open/close
- Filters
- Selected tab
- Upload progress
- Temporary form state

Do not put all application state into one global store.

---

55. API INTEGRATION

Frontend communicates with backend only through APIs.

Example:

```
Frontend
   ↓
HTTP API
   ↓
FastAPI
```

Do not:

- Put database credentials in frontend
- Call PostgreSQL directly from frontend
- Put AI API keys in frontend
- Trust frontend role information

---

56. RESPONSIVE BREAKPOINT BEHAVIOR

Desktop

Full sidebar + multi-column dashboards.

Tablet

Collapsible sidebar + reduced columns.

Mobile

Single-column layout.

Tables become:

```
Card
├── Primary information
├── Status
├── Important metadata
└── Actions
```

Document viewer becomes vertically stacked.

---

57. UX FLOW — PROCUREMENT OFFICER

```
Login
 ↓
Dashboard
 ↓
Create Tender
 ↓
Upload Tender
 ↓
AI Extracts Requirements
 ↓
Review Requirements
 ↓
Publish Tender
 ↓
Receive Bids
 ↓
Open Bid
 ↓
View Compliance
 ↓
Review Exceptions
 ↓
Generate Report
```

---

58. UX FLOW — BIDDER

```
Login
 ↓
Browse Tenders
 ↓
Open Tender
 ↓
Start Bid
 ↓
Upload Documents
 ↓
Document Validation
 ↓
Resolve Missing Documents
 ↓
Review Submission
 ↓
Submit
 ↓
Track Processing
 ↓
View Compliance Status
```

---

59. UX FLOW — REVIEWER

```
Login
 ↓
Review Dashboard
 ↓
Open Flagged Case
 ↓
Read Requirement
 ↓
Inspect Evidence
 ↓
Check Verification
 ↓
Make Decision
 ↓
Add Reason
 ↓
Submit
```

```
 ↓
Audit Event Created
```

---

60. UX FLOW — AUDITOR

```
Login
 ↓
Audit Dashboard
 ↓
Search Resource
 ↓
View Activity
 ↓
Open Decision History
 ↓
View Verification History
 ↓
Inspect Audit Trail
```

---

61. CORE UI SCREEN LIST

The MVP must include at least:

```
01. Login
02. Dashboard
03. Tender List
04. Create Tender
05. Tender Details
06. Requirement Management
07. Bid List
08. Bid Details
09. Bid Submission
10. Document Upload
11. Document Management
12. Document Viewer
13. Compliance Dashboard
14. Compliance Details
15. Verification Details
16. Review Queue
17. Review Workspace
18. Report Preview
19. Audit Logs
20. User Management
21. Profile
```

```
22. Settings
23. Notifications
```

---

62. DESIGN SYSTEM RULE

Build the frontend around a reusable design system.

Do NOT individually style every page.

Use shared:

```
Colors
Typography
Spacing
Buttons
Cards
Tables
Badges
Forms
Dialogs
Navigation
Status components
```

This ensures visual consistency.

---

63. IMPORTANT UX RULES

1. Never hide critical compliance evidence.
2. Never present AI recommendations as unquestionable facts.
3. Always show whether verification is real, unavailable, or mock.
4. Always distinguish AI result from human decision.
5. Use consistent status labels.
6. Make review-required cases highly visible.
7. Keep important actions within one or two clicks.
8. Do not expose unauthorized information.
9. Do not expose technical errors to normal users.
10. Preserve auditability of high-impact decisions.
11. Make document evidence easy to inspect.
12. Use clear empty/loading/error states.
13. Keep the interface professional and enterprise-oriented.
14. Avoid unnecessary animations.
15. Optimize the primary procurement workflow before adding decorative features.

---

64. FINAL UI/UX OBJECTIVE

The application should make a procurement officer immediately understand:

What does the tender require?

What did the bidder submit?

What information did the system extract?

What was verified?

Which requirements passed or failed?

Why did the system reach that result?

Which cases require human attention?

What is the final decision?

The complete experience should feel like a **secure AI-assisted procurement compliance workspace**, not simply an AI chatbot or document uploader.